

Confidence phenotypes: a unified computational account of value and decision certainty in reinforcement learning

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Abstract

Confidence, the “feeling of knowing” that accompanies every cognitive process, plays a critical role in human reinforcement learning; yet its computational bases in learning scenarios have only recently begun to be studied. Prior work has distinguished between value confidence (certainty in value estimates) and decision confidence (certainty that a choice is correct), but how these two forms of confidence are computed and interact has not been directly tested. Here we combine two experiments and previously published datasets to test competing computational hypotheses. We find that value confidence is best explained by a Bayesian computation reflecting the precision of value estimates, and that it adaptively guides behaviour by reducing exploration and promoting exploitation as certainty increases. In contrast, decision confidence departs from Bayesian predictions, especially on errors. A hybrid model integrating Bayesian probability with the precision of the decision variable better accounts for decision confidence. Moreover, the relative weights assigned to these two sources of information characterize individual differences in confidence reporting and, in addition, they are predictive of task and metacognitive performance, where the more Bayesian the subject, the higher the performance. Together, these results provide a unified computational mechanism through which distinct forms of confidence shape learning and choices in uncertain environments.

Keywords: reinforcement learning; decision-making; confidence; uncertainty.

Introduction

Humans routinely face uncertainty, about the state of the world, the outcomes of their actions, and the intentions of others. To navigate this uncertainty, the brain constructs internal beliefs and attaches to them a sense of confidence: a graded estimate of how reliable those beliefs are (Meyniel, Sigman, et al., 2015). Confidence, in this broad sense, permeates virtually every domain of cognition. For instance, it shapes how we learn from feedback (Meyniel, Schlunegger, et al., 2015), monitor and adjust decisions (Yeung & Summerfield, 2012), seek information (Desender et al., 2018), and coordinate with others (Bang et al., 2017). Its pervasive influence has led to the view that confidence is a central component of behavioural control (Schulz et al., 2023), and a key construct in transdiagnostic models of mental health (Hoven et al., 2019).

Despite its ubiquity, confidence is typically treated as a summary of how certain the brain is about its own states. This assumption has been highly productive, primarily through the use of perceptual paradigms and the study of decision confidence—the subjective probability of having made a correct choice (Figure 1; Fleming, 2024; Pouget et al., 2016). In this framework, confidence can bias subsequent choices (Lisi et al., 2020) or determine when to stop accumulating evidence (Balsdon et al., 2020; Balsdon & Philiastides, 2024), yet it is usually conceived as a more reflective quantity—a metacognitive evaluation of one's accuracy—rather than as a variable exerting a direct or instrumental influence on the decision process. In contrast, research in human reinforcement learning (RL) has centered on value confidence—the confidence or certainty in the values or expected outcomes associated with available options (Figure 1)—which plays a pivotal role in adaptive behavior by, for instance, governing the rate of belief updating, modulating the exploration–exploitation trade-off, and determining how new information is weighted against prior expectations (Behrens et al., 2007; Boldt et al., 2019; Nassar et al., 2010; Payzan-LeNestour & Bossaerts, 2011).

The contrast between the approaches used in perceptual and reinforcement-learning paradigms exposes a fundamental tension in how these two traditions conceptualize confidence: while perceptual models define it as a reflective readout of certainty, learning models reveal it as a generative signal that drives behaviour. How these roles coexist within a single computational architecture remains unknown—especially given that confidence itself is highly idiosyncratic (Navajas et al., 2017). Indeed, while individuals tend to report confidence consistently across time (Ais et al., 2016), their mappings between internal uncertainty and reported confidence vary widely across the population, suggesting that distinct computational strategies may underlie how uncertainty is evaluated and used to guide behaviour. Only a handful of studies have begun to address these questions in human RL (Boldt et al., 2019; Salem-Garcia et al., 2023; Ting et al., 2023), leaving open how the brain integrates confidence across the representational hierarchy—from beliefs about the world to confidence in the choices derived from them—and how it shapes decisions under uncertainty across individuals.

Here, we bring together these perspectives within a unified computational framework. Using both new and previously published reinforcement-learning datasets, we show that value confidence is well captured by the precision (i.e., the inverse of uncertainty) of Bayesian value estimates, whereas decision confidence departs from Bayesian predictions, reflecting a hybrid computation that integrates the Bayesian probability of being correct with the precision of the decision variable. The relative weighting of these components characterises individual computational profiles—'confidence phenotypes'—that predict performance, exploration strategies, and metacognitive accuracy. Together, these findings reveal that confidence is not a single computation but a family of mechanisms through which humans regulate learning and decision-making under uncertainty.

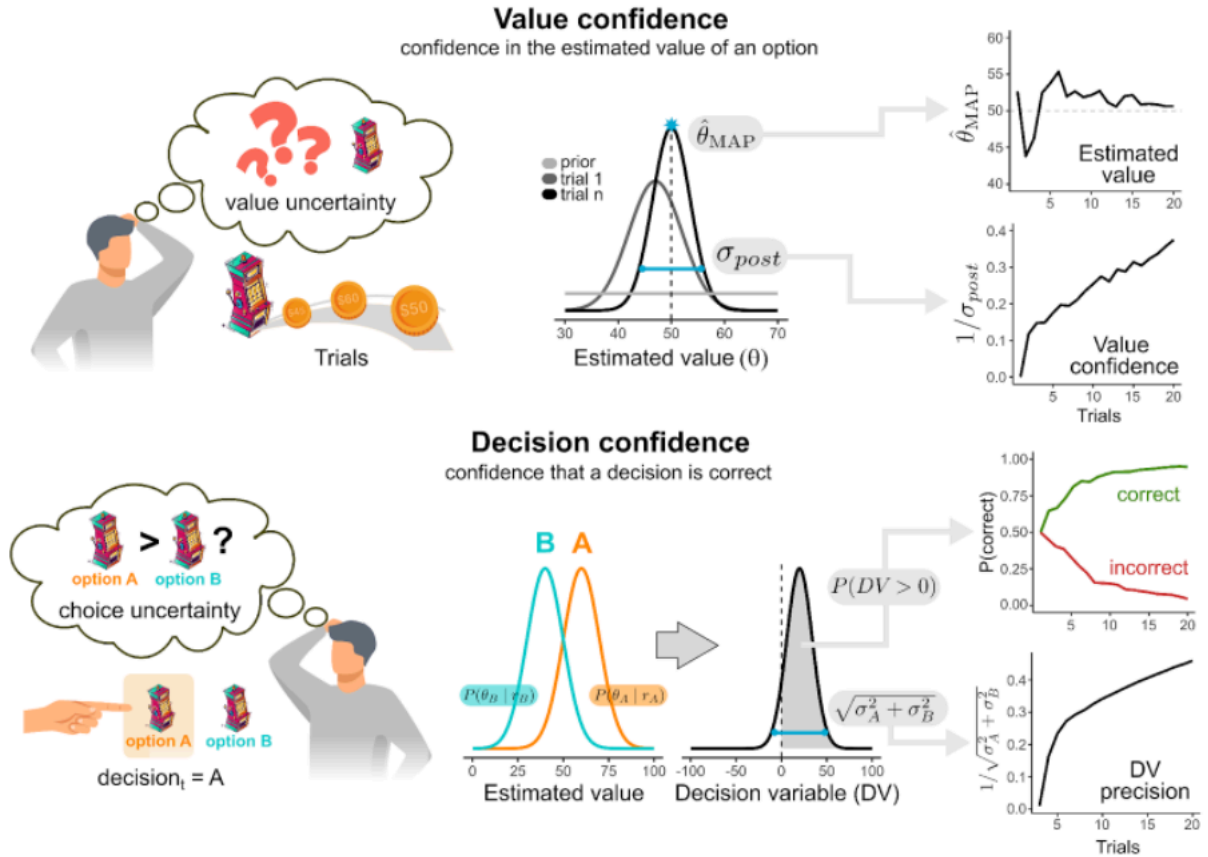


Figure 1 – Different levels of confidence in reinforcement learning contexts. Value confidence (top) represents the certainty about latent states of the environment, such as the value of one option in multi-armed bandit tasks. Under a Bayesian framework, value confidence can be straightforwardly modelled as the inverse of the standard deviation of a posterior distribution over the option’s estimated value (represented by θ in the figure). By observing subsequent rewards, not only the estimation of the option’s value gets more precise but the posterior shrinks, which is reflected in an increase in value confidence. Decision confidence (bottom) reflects the certainty of having made a correct choice. Under a Bayesian formulation, it can be understood as the probability of being correct, illustrated here by the area under the decision variable (the distribution obtained by subtracting the unchosen option’s posterior from the chosen option’s posterior) that it is greater than zero. As trials progress and certainty about the environment increases, the decision variable gets more precise and decision confidence better distinguishes between correct and incorrect responses. This increase in correct trials and decrease in incorrect trials is known as the “folded-x pattern” of confidence (Hangya et al., 2016; Sanders et al., 2016).

Results

Value confidence emerges from Bayesian inference

We began by modeling value confidence, i.e.: the confidence on the estimated values of the options. In the value confidence task from Boldt et al. (2019), on each trial participants ($N=21$) observed a reward randomly sampled from one of two arm-bandits. Then, they reported their belief on their mean value of that option and their confidence on this estimation. We tested several models to fit these confidence ratings: one model set was based on extensions of the Rescorla-Wagner (RW) algorithms, and the others were based on Bayesian inference (see Methods for details). Figure 2a depicts model fitting results of the best two models to this dataset (see Supplementary Figure 1 for the predictions of all tested models). We found that value confidence ratings were best explained by a Bayesian model (Model frequency (p_{model}) = 0.60, exceedance probability ($p_{exc.}$) > .99, protected exceedance probability ($p_{p.exc.}$) > .99; Figure 2c), in which value confidence reflected the inverse of the standard deviation of the posterior distribution over the estimated value of the option at play.

To test whether the Bayesian account of value confidence generalizes beyond the specific conditions of the Boldt et al. (2019) task, we next applied our models to data published by Quandt et al. (2022) (N=62 for their Experiment 1; N=60 for their Experiment 2). In their task, participants saw a sequence of a hundred rewards from one option in rapid presentation and then, as in Boldt et al. (2019), they had to report their estimated mean value of the option and their confidence on this estimate. A key manipulation was present in this dataset, namely that the variance of the reward distributions were different across alternatives, thus generating different levels of certainty across options which significantly affected value confidence judgments. This key manipulation allowed us to even better differentiate between our models, as some of the candidate models do not take the variance of the reward distributions into account, thus predicting a constant level of confidence regardless of the variance of the rewards (indeed, the inclusion of this dataset improved model recovery results, see Supplementary Figure 2). Specifically, we tested three models on Quandt et al. (2022) data: the mentioned Bayesian model, a RW model where value confidence reflected the square root of the number of rewards seen of the particular option at play (as it was the best non-Bayesian model in Boldt et al. dataset; also note that, in this dataset, this model make the same predictions as the other models that were a function of the number of rewards seen) and a RW model with a separate learning algorithm that tracked the variance of the rewards (as this one was the only RW model that had information about the variance of the rewards). Note that we did not include any of the models which involved a surprise term (nor the model where value confidence explicitly reflected the surprise of the reward seen, see Methods) as these models could not account for the data in the Boldt et al. (2019) dataset. We again found that the Bayesian model was the best fitting model (Exp. 1: $p_{model} = .94$; $p_{exc.} > .99$; $p_{p.exc.} > .99$; Exp. 2: $p_{model} = .96$; $p_{exc.} > .99$; $p_{p.exc.} > .99$; Figure 2b and Figure 2d), as it was able to capture the decrease in value confidence levels as the standard deviation of the options reward distributions increased (Figure 2b).

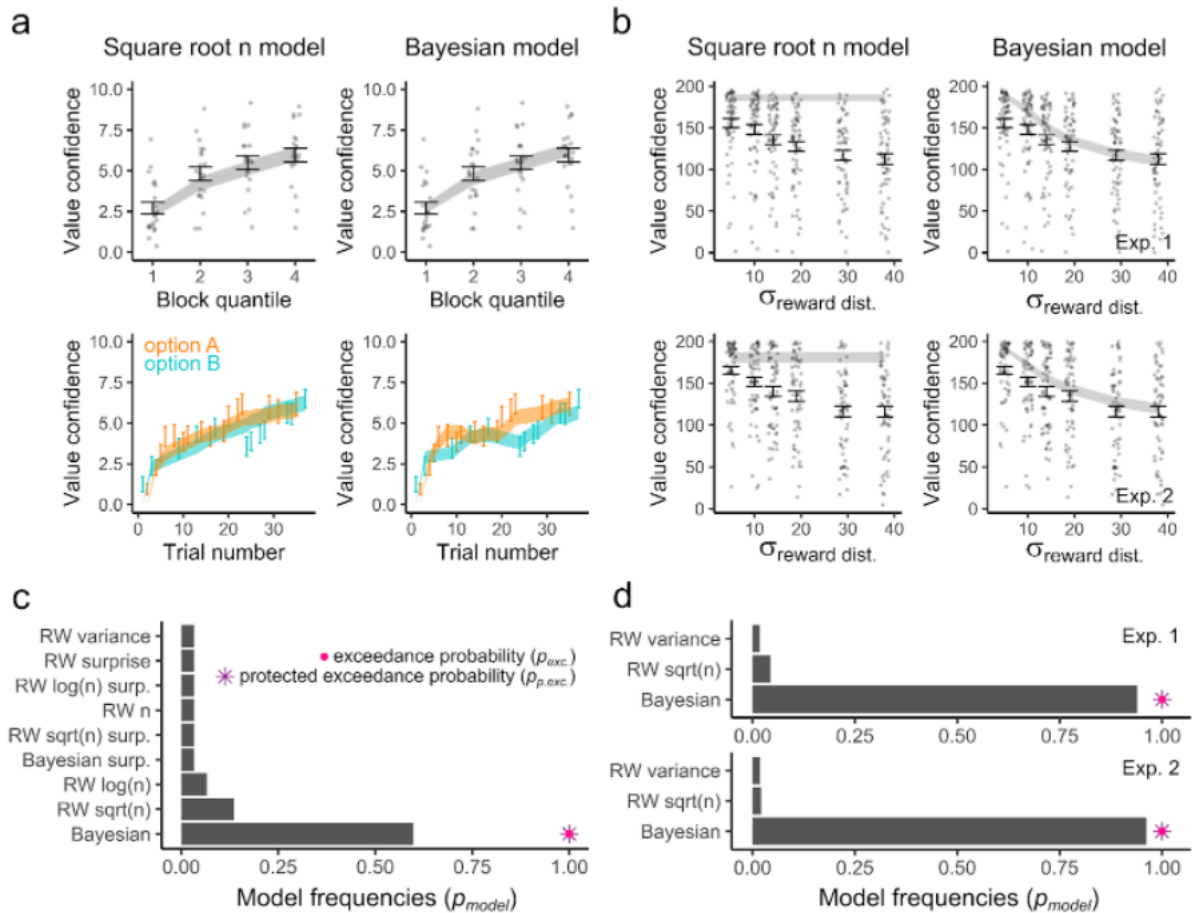


Figure 2 – Value confidence is captured by a Bayesian model. (a) Model fitting results for the two best models in the Boldt et al. (2019) data: the Bayesian model, where value confidence reflects the inverse of the standard deviation of the posterior distribution over an option's inferred value, and the Rescorla-Wagner model, where value confidence equals to the square root of the number of rewards observed of a specific option. First row:

models' fits to all the data. As blocks had different lengths, we divided blocks in four quantiles with respect to the trials to be able to pool all blocks together (x axis). Second row: example of models' fits in one block only. (b) Model fitting results to the Quandt et al. (2022) data. First row: models' fits to Exp. 1 data. Second row: models' fits to Exp. 2 data. Models that are a function of the number of rewards experienced (such as the RW sqrt(n) model) cannot account for the negative effect that the spread of the reward distribution has on value confidence. (c) Model comparison results. The Bayesian model was the best fitting model in Boldt et al. (2019) data. (d) Model comparison results. The Bayesian model was the best fitting model in both experiments in Quandt et al. (2022) data. In panels (a) and (b) error bars represent the standard error of the mean (SEM) of the behavioral data, shaded regions represent the SEM of models' predictions and dots represent individual averages.

Value confidence modulates the exploration-exploitation trade-off

A central functional role proposed for confidence is to regulate the balance between exploration and exploitation during learning (Boldt et al., 2019). Indeed, it is reasonable for an agent learning from the environment to increase exploitation as it becomes more certain about the values of the options at play. To test whether value confidence serves this adaptive control function, we extended the Bayesian model to include a parameter, b_1 , that modulates decision noise as a function of value confidence (see Methods section and Figure 3a). A positive b_1 indicates that decisions become more deterministic—that is, more exploitative—as value confidence increases. For testing this idea, we leveraged on the decision data from Boldt et al. (2019) experiment 2 study (N=30), and in two new studies conducted in our laboratory (N=29 and N=30; the latter a pre-registered replication). In all cases, participants performed a classic two-armed bandit task in which they had to choose between two options and rate their confidence on having selected the best one. After that, they saw the reward obtained.

To assess whether value confidence modulates the exploration–exploitation balance, we compared the Bayesian model including the b_1 parameter against the same model not including the parameter i.e., a model with constant decision noise. We found that the model including b_1 better explained the decision data in all datasets (Boldt et al. dataset: $p_{model} = .96$; $p_{exc.} > .99$; $p_{p.exc.} > .99$; Study 1: $p_{model} = .82$; $p_{exc.} > .99$; $p_{p.exc.} = .98$; Study 2: $p_{model} = .85$; $p_{exc.} > .99$; $p_{p.exc.} > .99$). Note that the main divergence between models emerged toward the end of each block, in line with the increase of exploitation as value confidence accumulates (Figure 3b, first and second rows). We also found that the b_1 parameter was significantly greater than zero in the three datasets employed (Boldt et al. dataset: $t_{29} = 8.76$; $p < .001$, $d = 1.60$; Study 1: $t_{28} = 3.95$; $p < .001$, $d = 0.73$; Study 2: $t_{29} = 5.58$; $p < .001$, $d = 1.02$; Figure 3b, third row), further validating the proposed mechanism.

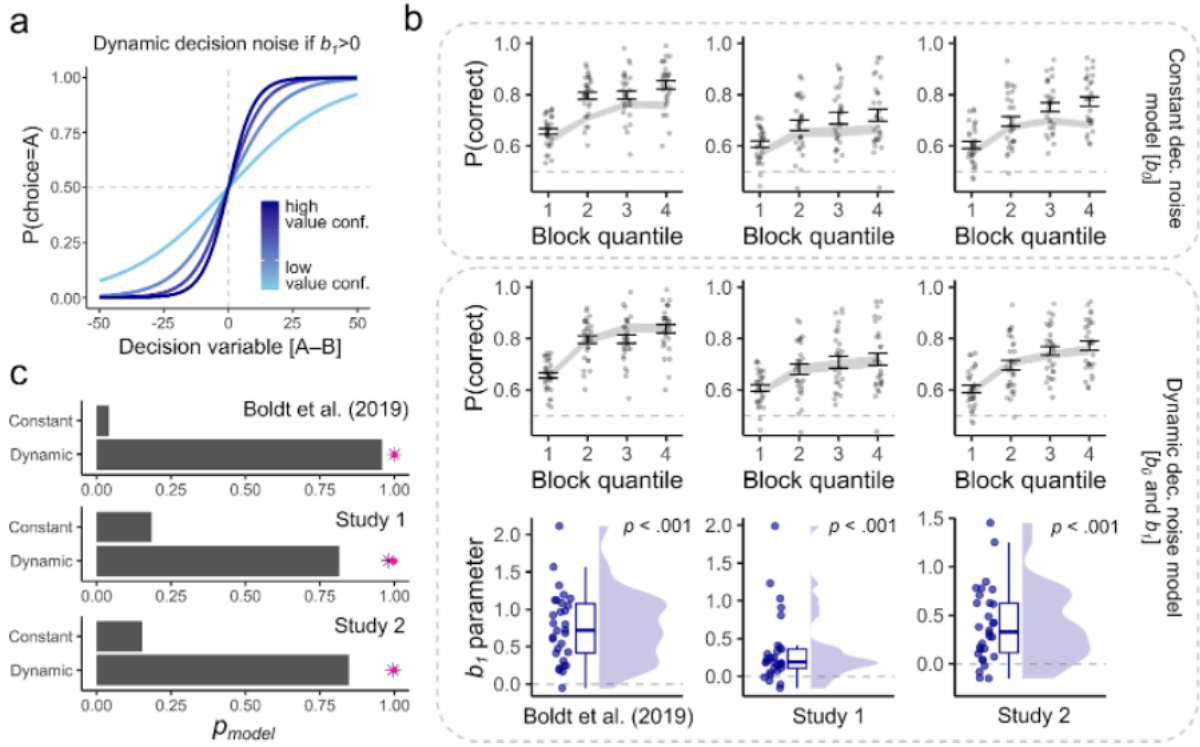


Figure 3 – Value confidence modulates the exploration-exploitation trade-off. (a) Illustration of an agent's behaviour if the b_1 parameter is positive. In such a case, as value confidence increases decision noise decreases because the slope of the softmax increases. This results in more exploitative decisions as value confidence increases throughout the trials. (b) Model fitting results of the models without (first row) and with (second row) dynamic decision noise (i.e.: without and with the b_1 parameter). Dashed lines in these two rows represent chance level performance (i.e.: a proportion of correct choices equal to .5). The third row depicts the distribution of the b_1 parameter across the three datasets analysed. Regarding the columns, the first column represents Boldt et al. data, the second column the Study 1 data and the third column the Study 2 data. (c) Model comparison results. The model that includes the modulation of decision noise by value confidence was the best fitting model across the three datasets. The conventions in this figure are equal to the ones in Figure 2.

Decision confidence deviates from the probability of being correct

Having established that value confidence and choice behavior are well described by Bayesian inference, we next evaluated whether the same Bayesian model could also explain *decision confidence*—the subjective certainty that a chosen option is correct. According to the Bayesian confidence hypothesis (Meyniel, Sigman, et al., 2015; Sanders et al., 2016), confidence should reflect the probability of being correct. Interestingly, while this model was able to capture the pattern of confidence in correct trials ($r_{354} = 0.87$; $p < .001$), it failed to account for confidence in incorrect trials ($r_{354} = 0.74$; $p < .001$; Figure 4b and Figure 4c, top row). Indeed, the Bayesian model predicts that confidence should progressively increase across trials for correct responses but decrease for the incorrect ones, a signature known as the “folded-x pattern” of Bayesian confidence (Hangya et al., 2016; Sanders et al., 2016). Inspired by Navajas et al. (2017), in which a similar deviation of Bayesian confidence in incorrect trials was found but in categorical decisions, we constructed an Bayesian-hybrid model, where confidence reflects a weighted combination of the probability of being correct (i.e.: the Bayesian confidence computation) and the precision of the decision variable. This model was able to capture decision confidence data both in correct ($r_{354} = 0.93$; $p < .001$) and incorrect trials ($r_{354} = 0.90$; $p < .001$; Figure 4b and Figure 4c, bottom row), and provided the best fit in two out of the three evaluated datasets (Boldt et al.: $p_{model} = 0.5$; $p_{exc.} = 0.5$; $p_{p.exc.} = 0.5$; Study 1: $p_{model} = 0.9$; $p_{exc.} > 0.99$; $p_{p.exc.} > 0.99$; Study 2: $p_{model} = 0.81$; $p_{exc.} > 0.99$; $p_{p.exc.} > 0.99$; Figure 4d). Across the three studies, in 65 out of 89 participants the hybrid model was preferred. We also evaluated a family of models that ignore option uncertainty and rely solely on estimated option values (as in Desender & Verguts, 2024; Salem-Garcia et al., 2023). A confidence model incorporating chosen option's value emerged as a reasonable alternative, but it failed to capture behaviour in bandit

tasks where both the mean and variance of reward distributions are manipulated (see Methods & Supplementary material).

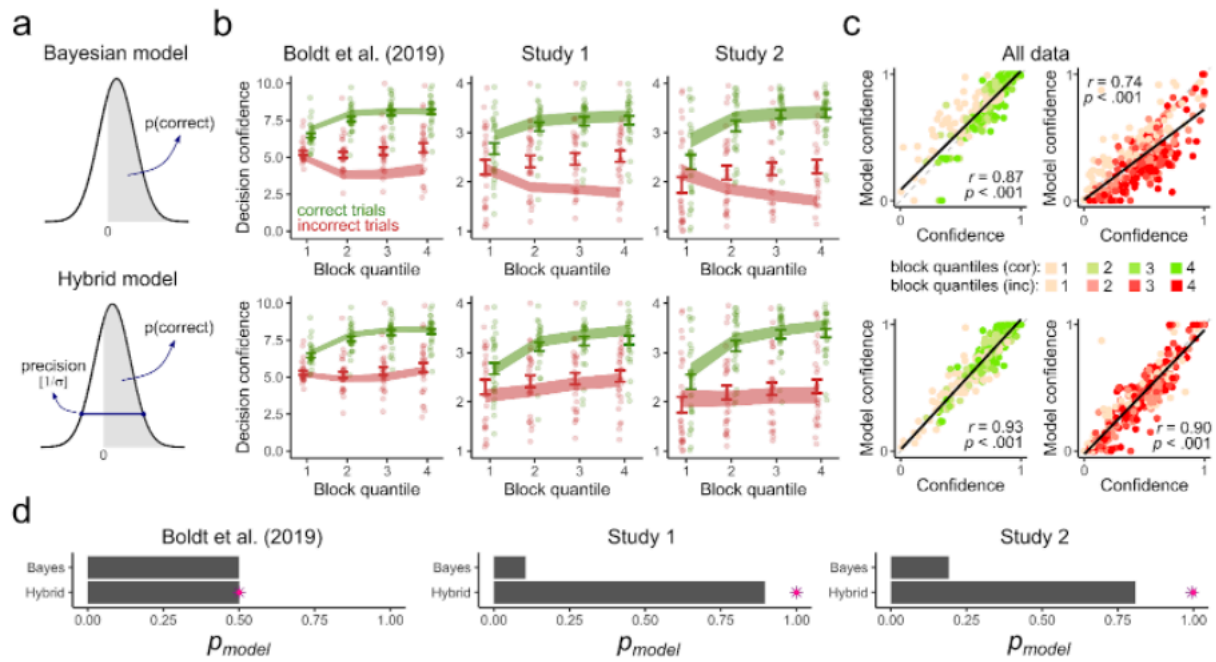


Figure 4 – Decision confidence deviates from the probability of being correct. (a) Schematic representation of the models. The subtraction between the two options’ expected values posterior distributions (i.e.: chosen posterior – unchosen posterior) creates a decision variable. In such a case, the probability of being correct is the area of this distribution that is greater than zero. This probability of being correct represents the Bayesian confidence (top panel). The Bayesian-hybrid model (lower panel) combines this probability with the precision (i.e.: the inverse of the standard deviation) of this decision variable distribution. (b) Model fitting results. The Bayesian model can capture correct trials but its prediction deviates from the data specially in incorrect trials (top-row). The Bayesian-hybrid model, on the other hand, can account for both correct and incorrect trials (bottom-row). (c) Correlations between models’ predictions and the data. Both models’ predicted confidence and empirical confidence data were rescaled to a range from 0 to 1 before computing the correlations (as confidence scales differed between datasets). The same pattern is found: the Bayesian model predictions deviate from the data specially in incorrect trials, where the correlation is considerably diminished. (d) Model comparison results. The hybrid model was the best fitting model in two out of three datasets. Considering all the data, the hybrid model was the preferred model for 65 out of 89 participants.

Individual signatures of decision confidence reveal distinct behavioural profiles

Having established a population-level deviation from Bayesian confidence, we next examined whether individuals differ systematically in how they combine the probability of being correct and the precision of the decision variable when reporting confidence. Following Navajas et al. (2017) we ran ordered linear regressions for each participant predicting decision confidence using the mentioned latent variables from the Bayesian-hybrid model. This yielded two regression coefficients—one for $p(\text{correct})$, $\beta_{p(\text{correct})}$, and one for precision, $\beta_{\text{precision}}$ —quantifying each individual’s reliance on these sources of information (Fig. 5a).

We then asked whether these computational weights were related to task and metacognitive performances. First, we ran a linear regression predicting task performance using both $\beta_{p(\text{correct})}$ and $\beta_{\text{precision}}$ as well as their interaction. We found that higher $\beta_{p(\text{correct})}$ values were predictive of higher performance ($\beta = 0.053$; $p < .001$; Figure 5b, left), but $\beta_{\text{precision}}$ values were not ($\beta = -0.034$; $p = .233$; Figure 5b, right). No interaction was found between the two predictors ($\beta = -0.009$; $p = .538$). Second, using a linear regression again, we evaluated whether these beta values were associated with the b_1 parameter values (i.e., the parameter that controlled the modulation by value confidence of the

exploration-exploitation trade-off). We found that higher $\beta_{p(\text{correct})}$ values were associated with higher b_1 parameter values ($\beta = 0.313$; $p < .001$; Figure 5c, left), while $\beta_{precision}$ values were not ($\beta = 0.082$; $p = .564$; Figure 5c, right), and no interaction was found between the two predictors ($\beta = -0.056$; $p = .442$). Finally, for evaluating metacognition—the ability to distinguish between correct and incorrect decisions—we run logistic regressions on each participant predicting accuracy from confidence judgments. We found that, as expected, larger $\beta_{p(\text{correct})}$ values were associated with higher metacognition ($\beta = 1.056$; $p < .001$; Figure 5d, left), but no relationship was found between metacognition and $\beta_{precision}$ ($\beta = 0.338$; $p = .104$; Figure 5d, right). This time, however, a negative interaction was found, suggesting that at higher values of $\beta_{precision}$ the positive effect of $\beta_{p(\text{correct})}$ diminished ($\beta = -0.224$; $p = .035$). Together, these results indicate that variability in confidence computation reflects distinct behavioral phenotypes rather than mere idiosyncrasies in reporting. Indeed, participants who relied more heavily on Bayesian computations of confidence tended to make more accurate decisions (Figure 5b), used their value confidence more effectively to guide exploitative choices (Figure 5c), and showed greater accuracy in evaluating the correctness of their decisions (Figure 5d). In contrast, greater reliance on precision computation proved suboptimal: it impaired metacognitive accuracy while exerting no detectable influence on performance exploration-exploitation strategies.

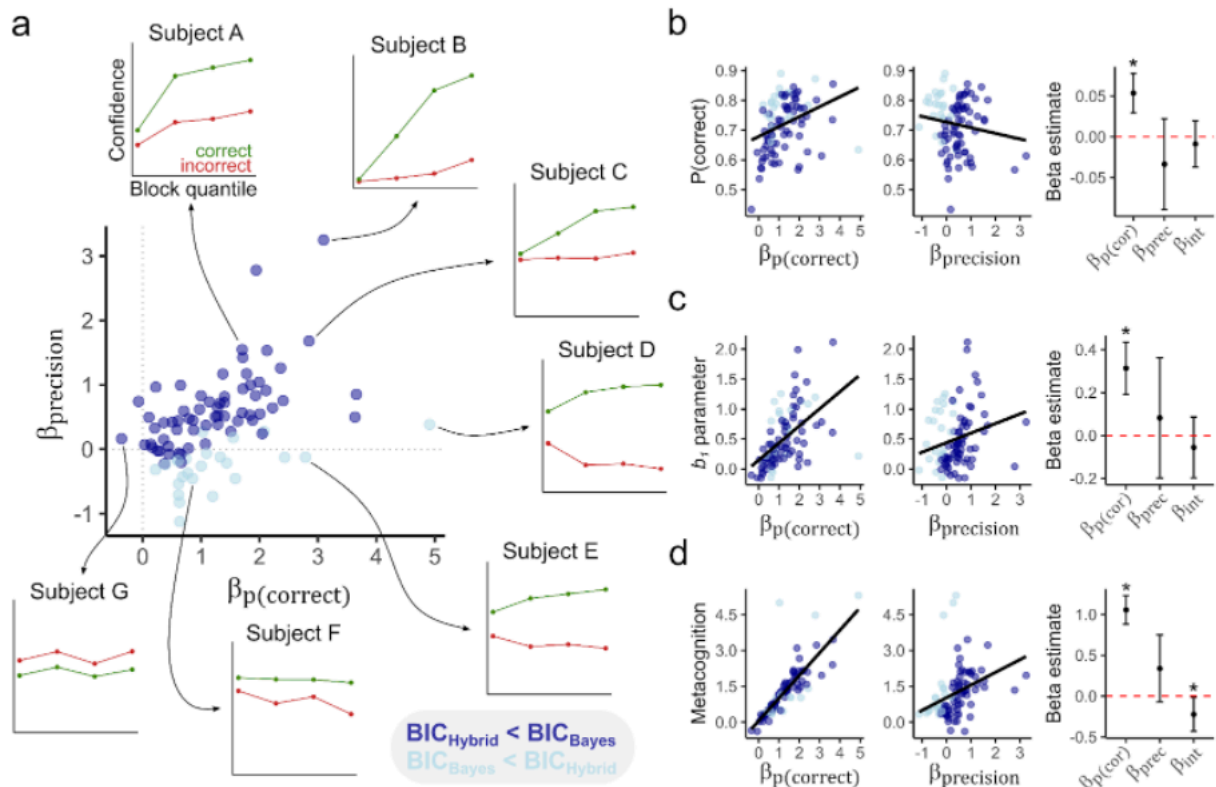


Figure 5 – Interindividual differences on confidence judgments predict task performance. (a) We found substantial individual variation in the way decision confidence was reported. By running a regression using the latent variables from the model, that is the probability of being correct and the precision of the decision variable, we were able to quantify this variation. Here we plot the beta values associated with the mentioned latent variables: $\beta_{p(\text{correct})}$ and $\beta_{precision}$. Participants that had both betas with positive values showed a pattern where confidence tended to increase in both correct and incorrect decisions (Participant A and Participant B), or not decrease in incorrect decisions (Participant C). Participants with near zero $\beta_{precision}$ values had a more Bayesian style of confidence reporting (illustrated by Participant D and Participant E). Interestingly, several participants had negative $\beta_{precision}$ values. At first value, this is surprising as an increase in precision should intuitively lead to greater confidence. However, the Bayesian model was the best model for virtually all of these participants (lightblue dots; note for instance the pattern of confidence judgments for Participant E), pointing out that the precision variable here should not be contributing and therefore its negative value suggests overfitting of the regression model. Note, however, that for some of these participants, like Participant F, a negative value appears to be justified as indeed her confidence decreased throughout the trials, that is, as precision increases.

Finally, some participants had negative $\beta_{p(\text{correct})}$ values, which consequently led to a poor metacognitive ability as incorrect decisions were associated with higher confidence (Participant G). (b) Increasing values of the $\beta_{p(\text{correct})}$ predicted higher task performance. (c) As with task performance, the $\beta_{p(\text{correct})}$ values predicted the values of the b_1 parameter (the parameter that controlled the exploration-exploitation trade-off with increasing value confidence levels). (d) As in the b_1 parameter case, only $\beta_{p(\text{correct})}$ values were associated with the metacognitive ability of the participants, that is, their ability to distinguish their correct and incorrect decisions with their confidence judgments. Note, however, that in this case a negative interaction between the two betas were found, meaning that increasing values of $\beta_{\text{precision}}$ negatively affected the ability of $\beta_{p(\text{correct})}$ to predict metacognition.

Discussion

Overall, our results reveal how value and decision confidence arise and interact in human reinforcement learning. By quantifying inter-individual differences in these two forms of confidence computations we uncovered different behavioral patterns, where participants whose confidence computations more closely matched the Bayesian model showed higher task performance and metacognitive accuracy—in line with normative principles of decision-making.

Value confidence—the certainty in the estimated values of available options—was best captured by Bayesian computations, reflecting the precision of a posterior distribution over those values. This aligns with several findings in the literature where Bayesian models capture human learning remarkably well (Bounmy et al., 2023; Gershman, 2018; Kang et al., 2024; Meyniel, 2020; Meyniel, Schlunegger, et al., 2015). Our work further strengthens this notion by validating the predictions of these models not only against choice behavior but also against participants' explicit value confidence ratings, providing a more direct test of the model's internal variables. It should be noted, however, that human reinforcement learning is not always strictly normative. Perhaps the most prominent deviations from normative behaviour in human reinforcement learning are the positivity and the confirmation biases, weighting new evidence more heavily when it is associated with a positive valence or consistent with prior beliefs (Chambon et al., 2020; Lefebvre et al., 2017; Palminteri et al., 2017). These effects are usually modelled using separate learning rates for positive or confirmatory evidence, allowing for a greater update of positive and confirmatory information (Palminteri & Lebreton, 2022). Given that we focused on the computations underlying the uncertainty around the estimated values—rather than the estimated values per se—it remains to be tested whether including separate learning rates would further improve model fits. Interestingly, recent work has proposed that these apparent biases can themselves emerge under Bayesian principles (Godara, 2025)—which may align with their evolutionary value (Hoxha et al., 2025)—therefore being a possibility that Bayesian updating can suffice to explain these phenomena.

Extending the Bayesian framework to decision data, we found that value confidence acts as a key variable modulating the exploration-exploitation trade-off. Indeed, as certainty increased over trials, decision noise systematically decreased, resulting in more exploitative decisions (see Desender & Verguts, 2024, for a similar result but using decision confidence rather than value confidence as a modulatory signal). This aligns closely with Thompson sampling algorithms where an agent explores more if uncertainty is higher (Gershman, 2018). It should be noted, however, that human behavior sometimes departs from this principle. In directed exploration cases, for instance, humans tend to deliberately select more uncertain options (Abir et al., 2024; Gershman, 2018). For other choice scenarios where direct and random exploration are likely to be at play, hybrid algorithms that combine directed and random exploration better explain human behavior (Gershman, 2018; E. Schulz & Gershman, 2019).

Although our model does not aim to capture all possible effects of (un)certainty on learning and decision-making, it highlights how Bayesian frameworks—by naturally representing uncertainty through precision metrics—can be extended to encompass a wide range of certainty-driven influences on

behavior. For instance, Bayesian models have been used to explain how balancing the approach and avoidance of uncertainty can reduce the cognitive costs of exploration in a resource-rational manner (Abir et al., 2024), and how uncertainty modulates the use of simple decision heuristics that imperfectly exploit immediate rewards (Paunov et al., 2024). It is also worth mentioning that, outside human RL and Bayesian modelling, work in value-based decisions has shown that, by fitting certainty-informed drift-diffusion models to human choices, certainty about options' values modulates not only choices but response times (Lee & Usher, 2023), and that confidence provides a benefit signal for allocating cognitive resources during decision formation (Bénon et al., 2024; Lee & Daunizeau, 2021). Including such models that account for decision-level dynamics as a decision rule in RL contexts could provide even more insight into the mechanisms by which confidence guides decisions in uncertain environments.

Decision confidence, on the other hand, deviated from the Bayesian computations that accounted for value confidence ratings and decisions. This is inline with previous work showing deviations of normative principles in decision confidence in the perceptual domain (Comay et al., 2023; Li & Ma, 2020; Lisi et al., 2020; Miyoshi & Sakamoto, 2025; Xue et al., 2024) as well as in reinforcement learning tasks (Salem-Garcia et al., 2023; Ting et al., 2023). Specifically, confidence departed from Bayesian predictions specially on incorrect trials, violating the well-known “folded-x pattern” of confidence: with increasing evidence, confidence should increase for correct trials but decrease for incorrect ones (Hangya et al., 2016; Sanders et al., 2016)—although it should be noted that this pattern is not always a prediction of normative models of confidence (Adler & Ma, 2018; Rausch & Zehetleitner, 2019). Decision confidence was better captured by a weighted combination of two sources: the (Bayesian) probability of being correct and the precision of the decision variable (which reflects the overall level of value confidence). Following the work of Navajas et al. (2017) in categorical decisions, we showed that the relative contributions of these two sources of information characterized the inter-individual patterns of decision confidence, reinforcing the notion that confidence is constructed in an idiosyncratic manner. Furthermore, the idea that the certainty about the options' values has an influence in decision confidence has a parallelism in influential models of confidence in perceptual decisions where stimulus reliability plays a key role in confidence computations (Boldt et al., 2017; Hellmann et al., 2023; Rausch et al., 2018; Shekhar & Rahnev, 2024) pointing to a possible cross-domains mechanism (although see Brus et al., 2021; Quandt et al., 2022). Extending this view, recent work has proposed that confidence itself reflects a noisy estimate of decision reliability constrained by a higher-order “meta-uncertainty” about the precision of the decision variable (Boundy-Singer et al., 2022). Incorporating such second-order uncertainty into learning frameworks could provide a fruitful direction for future research, offering a computational account of how confidence variability arises not only from task-related uncertainty but also from uncertainty about one's own inferential precision.

Importantly, these idiosyncratic patterns were not merely descriptive variations in confidence computation, but reflected distinct behavioral profiles. Participants whose confidence computations more closely followed the Bayesian ideal—weighting the probability of being correct more heavily—achieved higher overall accuracy, relied more strongly on value confidence to regulate their exploration-exploitation balance, and exhibited superior metacognitive insight into their own decisions. This suggests that the extent to which individuals approximate Bayesian confidence principles is not just a matter of internal calibration, but a signature of more adaptive learning and decision-making strategies. In this line, given that alterations in confidence are predictable of symptoms across multiple psychiatric dimensions (Hoven et al., 2019, 2023), these results may point to a continuum between optimal, Bayesian-like confidence computations and the kinds of suboptimality that characterize clinical populations.

Beyond idiosyncratic biases, recent work in human RL has shown that confidence judgments are also shaped by systematic biases (Salem-Garcia et al., 2023). Humans tend to overestimate their accuracy—the well-known overconfidence bias (Baranski & Petrusic, 1994)—and to report higher confidence when seeking gains than when avoiding losses, a phenomenon termed the valence-induced confidence bias (Lebreton et al., 2018). These effects can be formally accounted for by an overweighting

of the learned value of the chosen option in the computation of confidence, suggesting that confidence judgments partially inherit biases originating in the learning process. Moreover, individual differences in the learning parameters underlying these value biases—such as confirmatory updating and outcome-context dependency—predict the magnitude of metacognitive biases, establishing a computational bridge between biased learning and biased confidence (Salem-Garcia et al., 2023). While the Bayesian model proposed here does not explicitly capture such value- (although see Godara, 2025) and valence-related biases, extended Bayesian models—incorporating for instance reward-context dependencies—could provide a fruitful avenue for future research, offering a unified computational account of how learned values, value certainty, and decision confidence jointly give rise to both adaptive and biased behavior.

Finally, given that this recent research has modelled decision confidence using estimated option values but, unlike our approach, without incorporating uncertainty in those estimates (Desender & Verguts, 2024; Salem-Garcia et al., 2023), we therefore tested this kind of models on our datasets, as they offer a plausible alternative account of confidence behaviour. Consistent with these studies, a model driven by the value of the chosen option—akin to the “positive evidence bias” widely reported in perceptual decision-making (Maniscalco et al., 2016; Peters et al., 2017; Zylberberg et al., 2012)—provided a reasonable fit (see Supplementary material for model fitting results). However, it deviated from human behaviour mainly on incorrect trials, similarly to the Bayesian model (although over-estimating rather than under-estimating confidence in those trials). In addition, this model struggled to explain confidence patterns in a bandit task where both the mean and the variance of the reward distributions were manipulated (see Supplementary Material). Together, these findings support the idea that uncertainty in value estimates plays a central role in shaping confidence in learning settings. More broadly, we believe that future research could exploit systematic manipulations of reward distributions to further dissociate the respective contributions of estimated uncertainty and chosen-option value to decision confidence.

By linking Bayesian principles of learning with metacognitive evaluations of choice, our results bridge two traditionally separate literatures—on reinforcement learning and decision confidence—into a unified account of behavior under uncertainty. Beyond their theoretical relevance, these insights may inform how confidence guides learning and exploration across domains, and how its miscalibration contributes to suboptimal decision-making strategies.

Methods

All data & scripts to reproduce the reported results, as well as the Supplementary material, are available online at: osf.io/8tex5.

Datasets description

Here we describe the nature of the datasets employed in our study. For datasets that have been previously published, we refer the reader to the original publications for a more detailed description of the tasks employed. For the newly collected datasets, participants were university students who provided informed consent prior to participation. These new experimental protocols were approved by the Ethics Committee of the Instituto de Investigaciones Psicológicas (UNC–CONICET).

Value confidence

For the value confidence analysis, we employed datasets from two previous studies. First, we used the data from the experiment 1 of Boldt et al. (2019) study. Participants (N=21) performed a two-armed bandit task in which, on each trial, they observed a reward randomly sampled from one of the options. After seeing the reward, participants reported their belief on the mean value of this alternative and their

confidence on this estimate on a continuous square scale. Rewards were drawn from Beta distributions with the following parameters:

$\{\alpha = 1; \beta = 3\}; \{\alpha = 2; \beta = 3\}; \{\alpha = 3; \beta = 3\}; \{\alpha = 3; \beta = 2\}; \{\alpha = 3; \beta = 1\}.$

Participants performed 600 trials divided in 15 blocks of different lengths, ranging from 20 trials to 60 trials. Twenty five % of the trials were “decision trials” where participants had to choose one of the two options and rate their confidence on having chosen the best one.

The second dataset comprised the two experiments carried by Quandt et al. (2022). Both experiments followed identical procedures, with 62 participants in the first one and 60 in the second. Participants observed a hundred rewards from one alternative and then rated their belief on the expected value of the option and their confidence in this rating. They performed five trials with each alternative, for a total of six alternatives. The rewards associated each alternative were drawn from Normal distributions with the following parameters:

$\{\mu = 80, \sigma = 15\}; \{\mu = 100, \sigma = 20\}; \{\mu = 120, \sigma = 30\}; \{\mu = 130, \sigma = 40\}; \{\mu = 150, \sigma = 10\};$
 $\{\mu = 160, \sigma = 5\}.$

Decision confidence

For the decision confidence analysis we used four datasets. First, we used the data from experiment 2 of Boldt et al. (2019) study. Participants (N=30) performed a two-armed bandit task in which they had to choose, on each trial, one of the options with the aim to maximize their rewards. After choosing, they had to report on a continuous scale their confidence in having selected the best alternative. The alternatives' rewards distributions were sampled from the same Beta distributions of the value confidence experiment. Participants performed 600 trials, divided in 15 blocks of different lengths. Block length ranged from 20 trials to 60 trials. In 25% of the trials participants did not make a choice but instead reported their estimated mean value of the options (“rating trials”). These trials were excluded from the analysis.

The second dataset corresponded to a study conducted by us in our laboratory (“Study 1”). Participants (N=29) performed 20 blocks of 30 trials of a two armed bandit task. After choosing, they had to report their confidence in having selected the best option on a 1 (not sure at all) to 4 (completely sure) scale. Rewards were sampled from the same Beta distributions of the Boldt et al. (2019) study. The third dataset (“Study 2”) was a pre-registered replication of this protocol (N=30). Pre-registration is available at: <https://doi.org/10.17605/OSF.IO/Z5TCA>.

We employed a fourth dataset (“Study 3”) to differentiate between models with and without including certainty in estimated values (results are reported in the Supplementary material, see also Computational models section for the description of the models). This dataset was similar as the previous ones, with participants (N=15) performing 20 blocks of 30 trials of a two armed bandit task and reporting confidence on a 1 to 4 scale after each choice. However, the reward distributions were manipulated differently in this task. Inspired by Hertz et al. (2018) design, rewards from each option were sampled from Gaussian distributions, one with higher mean than the other, thus constituting one “correct” and one “incorrect” option. The variances of each one could independently be high ($H = 25^2 = 625$) or low ($L = 10^2 = 100$). This resulted in a design with four experimental conditions (thus participants completing 5 blocks with each condition, randomly ordered in the experimental session): H-H, H-L, L-H, L-L, where the first letter indicates the variance of the correct (higher expected value) and the second indicates the variance of the incorrect (lower expected value) option. The mean of the correct option was sampled uniformly from the range {40; 60} and the mean of the incorrect option was the mean of the correct option minus 30. The difference of value between options, therefore, was always the same.

Computational models

Values and value confidence

Two main classes of models were tested to account for the pattern of the value confidence data. One class consisted of models based on Rescorla-Wagner (RW) algorithms (R. A. Rescorla & A. R. Wagner, 1972), and the other was based on Bayesian inference. For the RW models, expected values for the options were updated as follows:

$$V_t = V_{t-1} + \delta(r_t - V_{t-1})$$

Where V represents the value of the option, t indexes the trial number, δ is the learning rate parameter and r is the reward obtained.

For the Bayesian models, posterior probability distributions of the options' expected values were constructed using Bayesian inference. As mentioned, in the Boldt et al. datasets, rewards were drawn from Beta distributions. Therefore, a Bayesian observer computes, on each trial, a posterior distribution over the mean of a Beta distribution, represented by θ , by integrating the likelihood of the rewards given θ and the prior probability of θ , which we modelled as uniform. Formally:

$$P(\theta | r_{1:t}) \sim P(r_{1:t} | \theta) P(\theta)$$

The maximum-a-posteriori (MAP) value of this posterior is therefore the predicted expected value of the option on trial t , i.e., V_t . These Bayesian models were implemented using Stan (*Stan*, 2025; MCMC diagnostics are reported in the Supplementary material). A re-parametrization was used to infer the mean of a Beta distribution. Specifically, we estimated the parameters that govern the shape of a Beta distribution, α and β , and then obtained the mean of the Beta as:

$$\theta = \frac{\alpha}{\alpha + \beta}.$$

Given that in Quandt et al. (2022) datasets rewards came from Normal distributions, we inferred the options' expected values (θ) by leveraging on the Normal-Normal conjugate family (Johnson, 2022). The posterior of θ , therefore, was therefore computed as follows:

$$P(\theta | r_{1:100}) \sim N\left(\bar{p} \frac{\sigma^2}{n\pi^2 + \sigma^2} + \overline{r_{1:100}} \frac{n\pi^2}{n\pi^2 + \sigma^2}, \frac{\pi^2 \sigma^2}{n\pi^2 + \sigma^2}\right)$$

Where $\bar{p}$ represents the mean of the prior (which we set at zero), π^2 represents prior variance (which we set at 100, thus constructing an uninformative prior), $\overline{r_{1:100}}$ is the mean of the rewards seen and σ^2 is the variance of the rewards seen (i.e.: the likelihood variance). The subscript 1:100 represents that subjects faced a hundred rewards for each option.

Finally, for the dataset of Study 3, reported in the Supplementary material, Bayesian inference was implemented using a Kalman-filter following Gershman (2018) to recursively infer posteriors' mean (θ) and variances (σ^2) for each option. Formally:

$$\theta_{t+1} = \theta_t + \delta_t(r_t - \theta_t)$$

$$\sigma_{t+1}^2 = \sigma_t^2 - \delta_t \sigma_t^2$$

where the learning rate δ_t is given by:

$$\delta_t = \frac{\sigma_t^2}{\sigma_t^2 + \tau^2}$$

where τ^2 was set to the specific option's reward distribution variance. For all options initial means (θ) were set at zero and initial σ^2 at 100 (uninformative prior).

Given these two modeling frameworks for value updating (i.e.: the Bayesian inference framework and the RW framework), we compared several models for explaining value confidence. The Bayesian framework naturally offers a measure for the certainty associated to the inferred values of the options, which is the inverse of the standard deviation of the posterior distribution over the expected value of the option at play. Thus, under the *Bayesian model*, value confidence can be expressed as:

$$value\ conf = \frac{1}{\sigma_{P(\theta_t|r_{1:t})}}$$

Here σ represents the standard deviation of the posterior distribution over θ , the estimated value of the specific option at play.

In contrast, as RW algorithms do not have an explicit measure of certainty in the values associated with each alternative, we extended this approach with several possible mechanisms for computing value confidence. The first of these extensions, the *RW surprise model*, defines value confidence as the inverse of the absolute surprise of the outcome. Formally:

$$value\ conf = \frac{1}{|r_t - V_{t-1}|}$$

The second RW model comprised a separated RW rule to track the variance of the outcomes (*RW tracking variance model*, Hertz et al., 2018):

$$\tau_t = \tau_{t-1} + \gamma[(r_t - V_{t-1})^2 - \tau_{t-1}]$$

Here τ refers to the computed variance of the rewards observed, and γ is the variance learning rate. Value confidence under this model reflected the inverse of the tracked variance, i.e.:

$$value\ conf = \frac{1}{\tau_t}$$

For the third RW model, value confidence reflected the number of rewards observed of the specific option sampled (*RW n model*). Let n refers to that number; value confidence is represented as:

$$value\ conf = n$$

Next, based on the *RW n* model, we constructed two additional variants: In this models value confidence reflects the logarithm of the number of rewards experienced with the specific option at play (*RW log(n) model*) or the square root of the same number (*RW sqrt(n) model*).

Finally, we tested a weighted combination of the Bayesian model, the RW sqrt(n) model and the RW log(n) model with the surprise generated by the outcome. The *Bayesian-surprise model* can be expressed as follows:

$$value\ conf = w \frac{1}{\sigma_{P(\theta_t|r_{1:t})}} - (1 - w)|(r_t - \theta_{t-1})|$$

The *RW sqrt(n)-surprise model* can be then stated as:

$$value\ conf = w\sqrt{n} - (1 - w)|(r_t - V_{t-1})|$$

And the *RW log(n)-surprise model* is the same as above but with the logarithm of n instead of the square root. Note that the weights (w) were independent parameters for each model.

All the RW models were tested in the Boldt et al. (2019) dataset. In the Quandt et al. (2022) dataset we only tested the *RW sqrt(n) model* and the *RW tracking variance model* given that 1) the *RW sqrt(n) model* was the best non-Bayesian model in Boldt et al. (2019) data; 2) the *RW tracking variance model* was the only non-bayesian model including information of the reward distribution variance for value confidence, the key manipulation in Quandt et al. (2022) dataset. The *Bayesian model* was included in all datasets.

Decisions and decision confidence

The decision confidence experiments involved classic two-armed bandit tasks where participants chose between two alternatives (which we will represent as A and B) and then had to rate their confidence on having chosen the best option (see Datasets description section). Given that we found that value confidence was best explained by a Bayesian model, we used Bayesian inference to update the values of the options on each trial as described for the value confidence data. Bayesian inference for options' posterior means and variances on each trial was implemented in Stan for the Boldt et al. (2019), Study 1 and Study 2 datasets and with a Kalman-filter for Study 3 dataset, as described in the Values and value confidence section above.

For fitting decisions, we used a classic softmax equation:

$$P(d_t = A) = \frac{1}{1 + \exp(-DV_t \lambda)}$$

Here d represents the decision, DV represents the decision variable (which is $V_{t-1}^A - V_{t-1}^B$, note that V here is the maximum-a-posteriori value of each option), and λ is the slope of the sigmoid that controls the noise in the decision process as an inverse temperature parameter. We tested two variations for fitting decisions. In one of them λ was treated as a constant. In the other, λ dynamically varied across trials, being modulated by value confidence in the following way:

$$\lambda_t = b_0 + b_1 \frac{1}{\sqrt{\sigma_{At-1}^2 + \sigma_{Bt-1}^2}}$$

Where σ represent the standard deviation of the posteriors over the options' expected values. The term multiplying b_1 can be interpreted as a global value confidence, as it reflects the inverse of the combined uncertainty associated with the value estimates of both options. As long as b_1 is positive, value confidence will modulate the exploration-exploitation trade-off by reducing decision noise with increasing values of value confidence. Conversely, if b_1 is indistinguishable from zero, decision noise is constant across trials and participants do not take into account the options' uncertainty for making decisions.

For decision confidence we tested several models. In the first one, confidence followed the Bayesian confidence hypothesis (Meyniel et al., 2015; Sanders et al., 2016), i.e.: confidence reflects the probability of being correct (*Bayesian model*). As the posteriors over options' expected values have Gaussian shape, this probability can be computed as follows:

$$confidence = P(correct_t) = \Phi\left(\frac{V_{t-1}^{chosen} - V_{t-1}^{unchosen}}{\sqrt{\sigma_{chosen,t-1}^2 + \sigma_{unchosen,t-1}^2}}\right)$$

Where Φ is the standard cumulative Gaussian function. In the second model, the *Bayesian-hybrid model*, decision confidence reflected a weighted sum of the probability of being correct and the certainty of the decision variable (Navajas et al., 2017). Formally:

$$confidence = \omega \Phi\left(\frac{V_{t-1}^{chosen} - V_{t-1}^{unchosen}}{\sqrt{\sigma_{chosen, t-1}^2 + \sigma_{unchosen, t-1}^2}}\right) + (1 - \omega) \frac{1}{\sqrt{\sigma_{chosen, t-1}^2 + \sigma_{unchosen, t-1}^2}}$$

In the main manuscript we report the results of these two primary models. However, because prior work on confidence in bandit tasks has relied on models that do not incorporate uncertainty in value estimates (Desender & Verguts, 2024; Salem-Garcia et al., 2023), we sought to assess whether including the certainty of the decision variable in the Bayesian-hybrid model was indeed warranted. To do so, we compared its fit to that of models that rely solely on the estimated option values. Specifically, we tested two extra models. For one of them, confidence reflected the absolute difference between the estimated values of the options (*Mean difference model*), i.e.:

$$confidence = |V_{t-1}^{chosen} - V_{t-1}^{unchosen}|$$

For the other extra model, and given that the value of the chosen option have been identified as important in explaining decision confidence both in bandit tasks and in perceptual decision making (i.e.: the “positive evidence bias”, Zylberberg et al., 2012), confidence reflected a weighted sum of the absolute difference between the estimated values of the options and the value of the chosen option (*Mean difference + PEB model*). Formally:

$$confidence = \omega |V_{t-1}^{chosen} - V_{t-1}^{unchosen}| + (1 - \omega) V_{t-1}^{chosen}$$

Model fitting results of these two extra models are reported in the Supplementary material.

Model fitting and model comparison procedure

We fitted all models by maximizing the log-likelihood of the parameters given each subject data using the *optim* function in R with the simulated annealing method. Ten optimization routines with different starting points were run per subject. The probability of a specific decision was given by the softmax equations detailed above. For computing the probability of confidence ratings (both value and decision confidence) in the case of Boldt et al. (2019) datasets and our datasets, we used a set of confidence criteria, $\{c_1, \dots, c_{k-1}\}$, where k is the number of confidence ratings, that divided a Gaussian distribution with mean at the confidence predicted by the model and a standard deviation that was fitted to each subject (akin as observational noise, Nunez et al., 2024). The probability of a specific level of confidence is then the area under this Gaussian that is restricted by corresponding confidence criteria. Boldt et al. (2019) confidence data was discretized from 0 to 10 to be able to apply this procedure. For Quandt et al. (2022) data we applied a linear transformation for mapping models' confidence to participants' confidence. We computed the likelihood of the parameters by determining the proportion of transformed confidence predictions from the model that matched the reported confidence divided by the total number of predictions (we used this approach to reduce the number of parameters as there were only 30 trials per subject).

We compared all models using Bayesian model selection at the group level (Stephan et al., 2009). Specifically, we computed Bayesian Information Criterion weights (Wagenmakers & Farrell, 2004) for each subject and for each model as a proxy for model evidence (i.e., the belief that model m generated data from subject i) and then used the *bmsR* R package to compute exceedance and protected exceedance probabilities (the probabilities that participants were more likely to use a certain model to generate behavior rather than any other alternative model) which were used as the metric for model comparison. We report in the Supplementary material parameter and model recovery results.

Statistical analyses

We also carried out several model free statistical tests. We used one-sample t-tests against zero (two-sided) for testing whether the b_1 parameter was greater than zero in the three datasets employed (Figure 3b). For each subject we run ordinal regressions predicting decision confidence on each trial with the two model-derived variables: the probability of being correct and the precision of the decision variable. Each of these two terms were therefore associated with two beta values, $\beta_{p(\text{correct})}$ for the first term and $\beta_{precision}$ for the second term, which allowed us to characterize individual differences in confidence reporting (Figure 5a). We then used these beta values (as well as their interaction) to predict task performance (the proportion of correct choices), the b_1 parameter and the metacognitive ability of each participant using the following linear regression models (Figure 5b, 5c, 5d):

$$\begin{aligned} p(\text{correct}) &\sim \beta_{p(\text{correct})} + \beta_{precision} + \beta_{p(\text{correct})} * \beta_{precision} \\ b_1 &\sim \beta_{p(\text{correct})} + \beta_{precision} + \beta_{p(\text{correct})} * \beta_{precision} \\ \text{metacognition} &\sim \beta_{p(\text{correct})} + \beta_{precision} + \beta_{p(\text{correct})} * \beta_{precision} \end{aligned}$$

Metacognition was measured using a logistic regression per subject predicting a correct (1) or incorrect (0) response using the confidence level reported by the participant on each trial:

$$P(\text{response} = \text{correct}) = \frac{e^x}{1 + e^x}$$

Where $x = \beta_0 + \beta_1 * \text{confidence}$.

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